Physics Bridge Observables Release for HAOS-IIP

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Abstract

This paper freezes release `53.0` of HAOS-IIP as an external physics-bridge layer over the scalar-carrier artifacts through `52.5`. The bridge does not modify HAOS core, add dynamics, claim continuum ontology, or identify the scalar carrier with physical spacetime. Instead, it post-processes frozen scalar observables into physics-facing proxy rows with explicit `PASS`, `OPEN`, and `WATCH` status. The resulting bridge read contains nine passing proxies, three open proxies, and one watched proxy. The central clarification is the inverse-square response: the raw recoverability-gradient read remains open, with flux constancy CV `0.313916` and refinement drift `0.524900`, but the shell-native law-aware reconstruction passes with minimum Green fit $R^2 = 0.973633$, maximum $|slope + 2| = 0.003353$, maximum scaled-response CV `0.006976`, and refinement drift `0.008949`. This supports only a bounded shell-native inverse-square / Newtonian-like response proxy on the tested scalar carrier, not a Newtonian gravity claim.

Contents

1 Scope

Release `53.0` is a bridge release, not a core release.

It introduces:

- a deterministic post-processing script for physics-facing proxy observables;
- generated CSV, JSON, and Markdown summaries;
- a cautious dictionary from scalar-carrier measurements to comparison variables.

It does not introduce:

- HAOS core changes;
- continuum ontology;
- Einstein-Hilbert recovery;
- a conserved stress-energy theorem;
- arbitrary localized inhomogeneity closure;
- a physical gravity claim.

2 Bridge Construction

The bridge reads only frozen scalar artifacts:

- `data/scalar_kernel_graph_metric_field_latest.json`;
- `data/scalar_kernel_graph_recoverability_gradient_latest.json`;
- `data/scalar_kernel_graph_recoverability_gradient_shell_native_latest.json`;
- `data/scalar_kernel_graph_current_closure_shell_native_latest.json`;
- `data/scalar_kernel_graph_current_closure_inhomogeneity_latest.json`;
- `data/scalar_kernel_graph_current_closure_radial_disorder_native_flux_latest.json`.

The generator is:

`experiments/physics_bridge/physics_observables.py`

It writes:

- `experiments/physics_bridge/results/physics_observables.csv`;
- `experiments/physics_bridge/results/physics_observables.json`;
- `experiments/physics_bridge/results/physics_observables_summary.md`.

3 Frozen Observable Table

Observable	Physics-facing analogue	Value	Target	Status
metric positive	effective metric positivity proxy	`1.076790`	≥ 0.800000	PASS
metric anisotropy	local isotropy proxy	`0.013949`	≤ 0.020000	PASS
metric drift	continuum-limit sanity proxy	`0.002350`	≤ 0.030000	PASS

Observable	Physics-facing analogue	Value	Target	Status
radial alignment	radial response-direction proxy	`0.990921`	≥ 0.940000	PASS
raw inverse-square	raw inverse-square response-law proxy	`0.313916 / 0.524900`	$\leq$ 0.120000/ $\leq$ 0.080000	OPEN
raw power fit	power-law scaling proxy	`0.321965`	≥ 0.950000	OPEN
shell-native inverse-square	shell-native Newtonian-like response proxy	`0.973633 / 0.003353 / 0.006976 / 0.008949`	$\geq$ 0.970000/ $\leq$ 0.020000/ $\leq$ 0.010000/ $\leq$ 0.020000	PASS
clean current	conserved-current analogue proxy	`0.073898 / 0.112930 / 0.096598`	$\leq$ 0.100000/ $\leq$ 0.150000/ $\leq$ 0.100000	PASS
current tail error	transport residual proxy	`0.271771 / 0.117611`	$\leq$ 0.300000/ $\leq$ 0.130000	PASS
smooth co-deformation	smooth weak-field co-deformation proxy	`0.146454 / 0.984018`	$\leq$ 0.150000/ $\geq$ 0.950000	PASS
localized bump	localized matter-like excitation boundary	`0.512392 / 0.962771`	$\leq$ 0.150000/ $\geq$ 0.950000	OPEN
disorder aggregate	disorder-robust transport-law proxy	`0.091117 / 0.300365 / 0.127842`	$\leq$ 0.100000/ $\leq$ 0.370000/ $\leq$ 0.150000	PASS
disorder seed margin	seed-universal disorder margin proxy	`2/3`	`3/3`	WATCH

4 Inverse-Square Clarification

The most important `53.0` correction is not that inverse-square behavior is simply "closed." It is more precise:

> the raw recoverability-gradient reconstruction remains open, while the shell-native law-aware reconstruction closes on the tested scalar carrier.

The raw read fails because:

- worst flux constancy CV is `0.313916`, above the threshold `0.120000`;
- profile drift is `0.524900`, above the threshold `0.080000`;
- minimum power-fit R^2 is `0.321965`, below the threshold `0.950000`.

The shell-native read passes because:

- minimum Green fit R^2 is `0.973633`, above `0.970000`;
- maximum $|slope + 2|$ is `0.003353`, below `0.020000`;

- maximum scaled-response CV is `0.006976`, below `0.010000`;
- profile drift is `0.008949`, below `0.020000`.

This supports a bounded inverse-square-like shell response under the stated reconstruction. It does not yet support a general Newtonian force law, source dynamics, stress-energy coupling, or continuum gravity.

5 What Remains Open

The open and watched boundaries are:

- raw inverse-square closure remains open;
- raw power-law fit remains open;
- localized bump closure remains open;
- seed-universal disorder-native flux remains watched because one aggregate-passing case still has only `2/3` seed-level passes.

6 Next Practical Moves

The next bridge steps should remain external:

- add refinement-scaling tables for the bridge proxies;
- generate bridge plots for radial response and metric spectra;
- locate the bump-width threshold where localized deformation begins to behave like smooth radial deformation;
- only afterward test curvature-like or stress-energy-like proxies.

7 Conclusion

Release `53.0` turns the scalar-carrier stack into a more physics-facing test protocol without mixing it into HAOS core. The correct result is structured: metric-like, shell-native inverse-square-like, clean transport, smooth inhomogeneity, and aggregate mild-disorder proxies pass; raw inverse-square closure, localized bump closure, and seed-universal disorder closure do not fully pass. That separation is the value of the bridge.